

ARISS Clock - READ ME

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Ver 3.0.0

ARISS Clock was designed to support how we do things at the K6DUE ARISS telebridge station. We need to know our time, UTC, and school time (to follow a script at 3am!). The color changes help us be aware of critical events. The elapse time is useful to note delays in establishing contact or when there were severe signal drop outs. We have it up as a large window on one of the monitors so everyone in the room can see it and have situational awareness.

Version 3.0.0 was written for Python 3.x on Linux.

Should I upgrade to V3.0.0? Yes, it fixes critical timing errors.

QUICK START

- To see what the clock looks like, see the `Screenshots` folder.
- To get started, click on the green `<> Code` button and choose `Download ZIP`.
- All the needed files are in the `ARISS_Clock_V3.0.0` folder.
- Read the `README` file for further instructions.
- **Don't forget to install the the two font files (`.ttf`) if they are not on your system.** This makes sure the numbers are properly aligned in columns. See sample image below.
- Edit the `ARISS_Clock_config.txt` file with UTC Rise and Set times.
- To run the Python script, you will need to have `Python 3.x` installed on your computer. You may need to install some additional libraries.
- This will run on a Raspberry Pi 3B+ and newer.
- You can make some limited changes to the look with **command line options**. See details in the `README` below.
- The window can be resized. See details in the `README` below.
- See the `HELPER` file for additional help and trouble shooting if you encounter issues or if you are new to Python.
- There is **no longer** an `.exe` version for Windows. If you are Python savvy, the `HELPER` file include instructions to generate an executable for your OS.

NAME

ARISS_Clock - Simple readable large clock to support ISS passes in support of ARISS school contacts at ground station K6DUE.

SYNOPSIS

Python script `python3 ARISS_Clock.py [ -b ] [ -c ] [ -e ] [ -h ] [ -l ] [ -t ]`

Executable (if available) `ARISS_Clock [ -b ] [ -c ] [ -e ] [ -h ] [ -l ] [ -t ]`

DESCRIPTION

ISS Rise and Set times are set in the configuration file. As the predicted Rise time approaches the timer changes colors. At the predicted Rise time the Rise timer stops at zero and the Set and Elapse Time (ET) timers start. The Set timer changes colors as the predicted Set time is approached. At the predicted Set time the Set timer and ET timers stop.

There are several clocks that are displayed. A UTC clock is displayed first. Next, the Local Time (LT) zone is detected for the local time clock. Finally, there is a optional event local time clock. Often the telebridge station is in a different time zone from the event being supported. Knowledge of the local time at the event is useful in following along with the scripted program. The Event Time Zone (ETZ) UTC time zone offset is specified in the config file. This clock label is fixed as Event Local Time (ELT).

While this tool is primarily intended to support ARISS telebridge stations for scheduled ISS contacts, it could be useful for ARISS moderators, or even locally at the events themselves. It could also be useful for direct contacts.

COMMAND LINE OPTIONS

-b, -B, --BW

Force Rise/Set timers to use only black & white colors. Default is to use color for active timers.

-c, -C, --Color

Force background colors off. Default is color.

-e, -E, --Event

Turn OFF display Event Local Time (ELT) clock. Default is to display the clock.

-h, -H, --Help

Lists the command line options in terminal window, then exits. This will override all other command line options.

-l, -L, --Labels

Turn OFF the display of the timer and clock labels. Default is to display the labels.

-t, -T, --Top

Remove timers from the top of the display and move to bottom. Default is to have timers at the top and clocks on bottom.

If any option is invalid the program uses all the defaults.

EXAMPLES

```
python3 ARISS_Clock.py
```

 Run the script from python using look and feel defaults.

```
python3 ARISS_Clock.py -l
```

Run the script from python with labels not displayed.

```
python3 ARISS_Clock.py -l -t -b or ARISS_Clock.py -l -t -b
```

Run the script or executable from python with all command line options. Order does not matter. Can be upper and lower case.

OVERVIEW

Predicted Rise and Set times can be viewed in a separate window by clicking on the "ARISS Contact Clock" button. Rise and Set predicted date/times are displayed in local and UTC. These should be verified against the satellite tracking software. Edit the config file if incorrect and restart the ARISS Clock.

The Rise countdown timer only shows the hours, minutes, and seconds. If Rise is more than 24 hours away, the Rise will get to zero and roll over. If Rise and Set have already passed when the script is started, the Rise, Set, and ET timers will all show zero. The timers change color over time as Rise and Set are reached, unless the `-b` command line option was used.

Timer colors change based on the time matching the config file Rise and Set times.

Rise and Set timer colors change as follows:

- Rise timer starts off GREEN when active.
- Set timer and ET start off grayed out while Rise is not zero.
- Rise timer goes YELLOW at less than 6 minutes to go. Warning!
- Rise timer goes RED at less than 1 minute(s) to go. Red alert!
- Rise timer goes GRAY at zero. The contact has started.
- Set timer starts off YELLOW when active. Contact in progress.
- ET timer turns blue when it becomes active.
- Set timer goes RED at less than 1 to go. Red alert!
- Set timer goes GRAY at zero when Set is reached. Contact has ended.

Clock window can be resized. Fonts are scaled based on window width. To shrink, recommend adjusting the width first, then the height. To enlarge, recommend adjusting the height first, then the width.

Clocks or timers at the bottom of the display can be rolled up and hidden from view. Grab the bottom of the window and drag up. Expand the window to expose.

Rise time is checked to make sure it is before the Set time. If not, an error message comes up. For setting the Rise and Set times, the date matters. The Rise and Set timers will not change unless the date and time matches the UTC time clock. UTC is calculated based on the system's time zone information read from the operating system.

FILES AND DIRECTORIES

All the files should be in the same folder.

ARISS_Clock.py

Python script. Main program. Requires python 3.x to run. Requires a number of Python libraries that may not be included with Python by default. Use pip to install.

ARISS_Clock_config.txt

The Rise and Set times and school time zone are set in the configuration file. Instructions are included in the file. At startup the configuration file is read. If the config file is not found in the same folder as the program, a new config file is created and the Rise and Set times will need to be updated.

Verify configuration file version matches the ARISS Clock version. Older versions are not compatible with this version. If it does not match, then delete the configuration file and restart. A new file will be generated with default values Edit for new school time zone offset, Rise and Set date/times.

If the configuration file gets corrupted, just delete it, and restart the ARISS Clock. A new file will be created.

ARISS_Clock_readme.txt

This help file. The file is generated every time the program is started up.

ARISS_logo.png

ARISS logo image used by the python script. Must be present if running the .py file. Built into the executable.

ARISS_logo_simple.ico

ARISS logo icon image. Used by Windows.

MAKING AN EXECUTABLE

This package ONLY includes the Python scripts.

To run this under different operating systems, there are two choices:

1. Install Python and run from the Python environment
2. Generate an executable under your OS

It is possible to create executable for your OS with the steps listed below. This is not for the beginner. It has worked for Linux, Raspberry Pi, and Windows 10. It should work for the MacOS, I just don't have one to test with. Note that outside of Windows, binary executable are often not portable (i.e. a Linux binary will not run on a Raspberry Pi).

The process requires a Python 3.xx installation, `pyinstaller` script, and all the required Python libraries imported by `ARISS_Clock.py`. It also requires files `ARISS_logo_simple.png` and `ARISS_logo_simple.ico` files.

On Linux/Raspberry Pi use command line:

```
pyinstaller --onefile -w -F -i "ARISS_logo_simple.ico" --add-data 'ARISS_logo.png:.' 'ARISS_Clock.py'
```

Windows 10 pyinstaller command line:

```
pyinstaller -w -F -i "ARISS_logo_simple.ico" --add-data ARISS_logo.png;. --add-data ARISS_logo_simple.ico;. ARISS_Clock.py
```

The binary executable must reside in the folder with the other files from the package.

Once an binary executable has been made, a shortcut can be created. Note that in the shortcut you can specify the command line switches as desired.

FONTS

This script tries to use the "DejaVu Sans Mono" non-proportional font. Not all systems may have this font. If the specified font is not found it gets substituted with a different font and this will affect the look and feel. It is strongly recommend to install the correct fonts. The fonts are free. See INTERNET RESOURCES below.

Required font files:

- DejaVuSansMono.ttf
- DejaVuSansMono-Bold.ttf

AUTHOR

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INTERNET RESOURCES

DejaVu Fonts at <https://dejavu-fonts.github.io> or from 1001 Fonts at <https://www.1001fonts.com/dejavu-sans-mono-font.html>